Show all steps of your solution Final Answers with no clear solution get ZERO

*Note: work to 4 decimal places in all Questions*

**Q1**- Use the *Bairstow Method* applying first iteration, to find the roots of the polynomial: **(10 marks)**

$f(x) = x^4 - 3x^3 + 5x - 4$. Use $(x^2 - 2x - 1)$, as a starting factor. $r=2$ $s=1$

After the first iteration, the values for the second iteration are:

$r_{new}$ =3.6667.... $s_{new}$ =-4.3334....

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| | $x^4$ | $x^3$ | $x^2$ | $x$ | const |
|---|---|---|---|---|---|
| | 1 | -3 | 0 | 5 | -4 |
| 2 | ↓ | 2 | -2 | -2 | 4 |
| 1 | ↓ | | 1 | -1 | -1 |
| | 1 | -1 | -1 | 2 | -1 |
| | $b_0$ | $b_1$ | $b_2$ | $b_3$ | $b_4$ |
| 2 | ↓ | 2 | 2 | 4 | |
| 1 | ↓ | | 1 | 1 | |
| | 1 | 1 | 2 | 7 | |
| | $c_0$ | $c_1$ | $c_2$ | $c_3$ | |

$$\begin{bmatrix} 2 & 1 \\ 7 & 2 \end{bmatrix} \begin{bmatrix} \Delta r \\ \Delta s \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$\Delta r = 1.6667$

$\Delta s = -5.3334$

$r_{new} = 1.6667 + 2 = 3.6667$

$s_{new} = -5.334 + 1 = -4.3334$

10

**Q2**- Consider the following three simultaneous linear equations: (10 marks)

$$2x_1 + 2x_2 + x_3 = 4$$
$$3x_1 - 2x_2 + 2x_3 = 7$$
$$4x_1 + 3x_2 + \alpha x_3 = 10$$

Given that the equations have no solution, Use the ***Cramer's Rule*** to find the value of $\alpha$

$\alpha =$ 2.1

$$\det A = \begin{bmatrix} 2 & 2 & 1 \\ 3 & -2 & 2 \\ 4 & 3 & a \end{bmatrix} = 2\begin{bmatrix} -2 & 2 \\ 3 & a \end{bmatrix} - 2\begin{bmatrix} 3 & 2 \\ 4 & a \end{bmatrix} + \begin{bmatrix} 3 & -2 \\ 4 & 3 \end{bmatrix} = 0$$

$$2(-2a-6) - 2(3a-8) + 17 = 0$$

$$a = 2.1$$

**Q3-** Use the ***Golden Section Search Method*** to find the maximum of: **(10 marks)**

$f(x) = -0.5x^3 + sin(2x)$. Perform 3 iterations assuming: $x_l = -0.5, \quad x_U = 1$

| $x_l$ | $x_U$ | $d$ | $x_1$ | $x_2$ | $f(x_1)$ | $f(x_2)$ |
|---|---|---|---|---|---|---|
| -0.5 | 1 | 0.927o | 0.427o | 0.073o | 0.7149 | 0.1452 |
| 0.073 | 1 | 0.5729 | 0.6459 | 0.4271 | 0.8266 | 0.715o |
| 0.4271 | 1 | 0.354o | 0.7811 | 0.646o | 0.7616 | 0.8265 |

---

$d_1 = 0.61803(1 - -0.5) = 0.927o$

$x_1 = -0.5 + 0.927o = 0.427o$

$x_2 = 1 - 0.927o = 0.073o$

$f(x_1) = 0.7149 \quad x_L = 0.073$

$f(x_2) = 0.1452 \quad x_u = 1$

---

$d_2 = 0.61803(1 - 0.073) = 0.5729$

$x_1 = 0.6459$

$x_2 = 1 - 0.5729 = 0.4271$

$f(x_1) = 0.8266 \quad x_L = 0.4271$

$f(x_2) = 0.715o \quad x_u = 1$

---

$d_3 = 0.61803(1 - 0.4271) = 0.354o$

$x_1 = 0.4271 + 0.354o = 0.7811$

$x_2 = 0.646o$

$f(x_1) = 0.7616$

$f(x_2) = 0.8265$

*End of Questions*

**Show all steps of your solution Final Answers with no clear solution get ZERO**

***Note: work to 4 decimal places in all questions***

**Q1-** Use the ***Bairstow Method*** applying first iteration, to find the roots of the polynomial: **(10 marks)**

$f(x) = x^4 - 3x^3 + 5x - 4$. Use $(x^2 - 2x - 1)$, as a starting factor.

After the first iteration, the values for the second iteration are:

$r_{new}$ =……, $s_{new}$ =……..

---

| | | | | | |
|---|---|---|---|---|---|
| Coefficients | 1 | -3 | 0 | 5 | -4 |
| r=2 | | 2 | -2 | -2 | 4 |
| s =1 | | | 1 | -1 | -1 |
| ***b values*** | 1 | -1 | -1 | 2 | -1 |
| r=2 | | **2** | **2** | **4** | **14** |
| s =1 | | | **1** | **1** | **2** |
| ***c values*** | **1** | **1** | **2** | **7** | **15** |

**$b_1 = 2$** **$b_0 = -1$** **$C_2 = 1$** **$C_2 = 2$** **$C_1 = 7$**

**(5marks)**

$$\begin{cases} c_2\Delta r + c_3\Delta s = -b_1 \\ c_1\Delta r + c_2\Delta s = -b_0 \end{cases} \Rightarrow \begin{cases} 2\Delta r + 1\Delta s = -2 \\ 7\Delta r + 2\Delta s = 1 \end{cases}$$

$\boldsymbol{\Delta r = 1.6666}$

$\boldsymbol{\Delta s = -5.3333}$ **(3marks)**

$r_{\text{new}} = r + \Delta r = 2 + 1.66667 = \mathbf{3.6666}$

$s_{\text{new}} = s + \Delta s = 1 - 5.33333 = \mathbf{-4.3333}$ **(2marks)**

**Q2-** Consider the following three simultaneous linear equations: **(10 marks)**

$$2x_1 + 2x_2 + x_3 = 4$$

$$3x_1 - 2x_2 + 2x_3 = 7$$

$$4x_1 + 3x_2 + \alpha x_3 = 10$$

Given that the equations have no solution, Use the ***Cramer's Rule*** to find the value of $\boldsymbol{\alpha}$

$\boldsymbol{\alpha}$ = ……2.1…..……

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$$D = \begin{vmatrix} 2 & 2 & 1 \\ 3 & -2 & 2 \\ 4 & 3 & \alpha \end{vmatrix} = 0$$ **(3marks)**

$$D = 2\begin{vmatrix} -2 & 2 \\ 3 & \alpha \end{vmatrix} - 2\begin{vmatrix} 3 & 2 \\ 4 & \alpha \end{vmatrix} + 1\begin{vmatrix} 3 & -2 \\ 4 & 3 \end{vmatrix}$$ **(4marks)**

$$-10\alpha + 21 = 0$$

$$-10\alpha = -21$$

$$\alpha = \frac{21}{10} = 2.1$$ **(3marks)**

**Q3-** Use the ***Golden Section Search Method*** to find the minimum of: **(10 marks)**

$f(x) = -0.2x^2 + cos(2x)$. Perform 3 iterations assuming: $x_l = 2, \quad x_U = 4.5$

| $x_l$ | $x_U$ | $d$ | $x_1$ | $x_2$ | $f(x_1)$ | $f(x_2)$ |
|---|---|---|---|---|---|---|
| 2.0000 | 4.5000 | 1.5451 | 3.5451 | 2.9549 | -1.8679 | -0.8209 (3marks) |
| 2.9549 | 4.5000 | 0.9549 | 3.9098 | 3.5451 | -3.0448 | -1.8679 (3marks) |
| 3.5451 | 4.5000 | 0.5902 | 4.1353 | 3.9098 | -3.8555 | -3.0448 (4marks) |

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***End of Questions***

**Show all steps of your solution Final Answers with no clear solution get ZERO**

***Note: work to 4 decimal places in all questions***

**Q1-** Use the ***Bairstow Method*** applying first iteration, to find the roots of the polynomial: **(10 marks)**

$f(x) = 2x^4 - 3x^3 - 5x + 4$. Use $(x^2 - x - 2)$, as a starting factor.

After the first iteration, the values for the second iteration are:

$r_{new}$ =....., $s_{new}$ =.....

---

| | | | | | |
|---|---|---|---|---|---|
| Coefficients | 2 | -3 | 0 | -5 | 4 |
| r=1 | | 2 | -1 | 3 | -4 |
| s =2 | | | 4 | -2 | 6 |
| ***b values*** | 2 | -1 | 3 | -4 | 6 |
| r=1 | | **2** | **1** | **8** | **6** |
| s =2 | | | **4** | **2** | **16** |
| ***c values*** | **2** | **1** | **8** | **6** | **28** |

**$b_1 = -4$** **$b_0 = 6$** **$C_3 = 1$** **$C_2 = 8$** **$C_1 = 6$**

**( 5marks)**

$c_2\Delta r + c_3\Delta s = -b_1 \Longrightarrow 8\Delta r + 1\Delta s = 4$

$c_1\Delta r + c_2\Delta s = -b_0 \Longrightarrow 6\Delta r + 8\Delta s = -6$

$\boldsymbol{\Delta r = 0.6551}$

$\boldsymbol{\Delta s = -1.2413}$

**( 3marks)**

$r_{\text{new}} = r_{\text{old}} + \Delta r = 1 + 0.6552 = \mathbf{1.6552}$

$s_{\text{new}} = s_{\text{old}} + \Delta s = 2 - 1.2414 = 0.7586$

$\boldsymbol{r_{\text{new}} = 1.6552}$, $\boldsymbol{s_{\text{new}} = 0.7586}$ **( 2marks)**

**Q2-** Consider the following three simultaneous linear equations: **(10 marks)**

$$x_1 + 2x_2 + x_3 = 4$$

$$3x_1 - x_2 + 2x_3 = 7$$

$$4x_1 + x_2 + \alpha x_3 = 10$$

Given that the equations have no solution, Use the ***Cramer's Rule*** to find the value of $\boldsymbol{\alpha}$

$\boldsymbol{\alpha}$ = ……..……

---

$$D = \begin{vmatrix} 1 & 2 & 1 \\ 3 & -1 & 2 \\ 4 & 1 & \alpha \end{vmatrix} = 0$$ **(3marks)**

$$D = 1\begin{vmatrix} -1 & 2 \\ 1 & \alpha \end{vmatrix} - 2\begin{vmatrix} 3 & 2 \\ 4 & \alpha \end{vmatrix} + 1\begin{vmatrix} 3 & -1 \\ 4 & 1 \end{vmatrix}$$ **(4marks)**

$$-7\alpha + 21 = 0$$

$$-7\alpha = -21$$ **(3marks)**

$$\alpha = 3$$

**Q3-** Use the ***Golden Section Search Method*** to find the maximum of: **(10 marks)**

$f(x) = -0.5x^3 + sin(2x)$. Perform 3 iterations assuming: $x_l = -0.5, \quad x_U = 1$

| $x_l$ | $x_U$ | $d$ | $x_1$ | $x_2$ | $f(x_1)$ | $f(x_2)$ |
|---|---|---|---|---|---|---|
| -0.5000 | 1.0000 | 0.9271 | 0.4271 | 0.0729 | 0.7150 | 0.1452 (3marks) |
| 0.0729 | 1.0000 | 0.5730 | 0.6459 | 0.4271 | 0.8265 | 0.7150 (3marks) |
| 0.4271 | 1.0000 | 0.3541 | 0.7812 | 0.6459 | 0.7616 | 0.8265 (4marks) |

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***End of Questions***